

Siyuan Qian

📍 Beijing, China 📞 +86-135-0139-3580
✉ qiansiyuan@stu.pku.edu.cn ✉ siriyp@gmail.com
🌐 Homepage 🔄 SiriYep 🎓 Google Scholar



Profile

Ph.D. candidate in the HMI Lab, School of Computer Science, Peking University, advised by Assistant Professor Shanghang Zhang. My research develops generalizable multimodal agents that perceive, reason, and act across complex physical and virtual environments. I am particularly interested in vision-language-action (VLA) models, world models, and hierarchical architectures for long-horizon decision-making, with applications spanning real-world mobile manipulation, complex multi-stage robotic tasks, and decision-making for game agents. My publications at NeurIPS, RSS, ACM MM, and *Nature Computational Science* have received 300+ Google Scholar citations; I also received a Best Paper Award at the RSS 2026 Embodied4Arts Workshop and have served as a reviewer for NeurIPS 2025, ICML 2025/2026, and ECCV 2024/2026.

Education

Ph.D.	Peking University , School of Computer Science, Computer Applied Technology Research: multimodal perception and adaptive action decision-making for embodied intelligence Focus: Embodied AI, VLA, world models, and VLMs	2022.09 – 2027.06
B.Eng.	Beihang University , School of Electronic Information Engineering Honors: highest undergraduate honor in Beihang’s Information Science Division (top 0.3%); highest undergraduate honor in the School of Electronic Information Engineering (top 3%); University Outstanding Graduate (top 10% in major)	2018.09 – 2022.06

Internship Experience

miHoYo , Game AI Research Intern	Beijing & Shanghai, China 2026.07 – Present
- Contributed to a hierarchical high-level/low-level fast-slow architecture: a deliberative module handles semantic understanding, long-horizon planning, and complex decisions, while a reactive module provides low-latency action selection and execution, balancing generalization, reasoning, and runtime responsiveness. - Systematically reviewed external data samples, co-developed multiple versions of data-collection and quality-acceptance standards, and validated the first delivered batch. - Researched structured representations of game scenes and assets and developed a scene representation encoder that converts structured information into multimodal-model inputs, exploring deployment without an additional agent-facing RGB rendering pass. - Built an initial evaluation environment and workflow for assessing model behavior, action execution, and deployment feasibility in representative scenarios.	
Beijing Innovation Center of Humanoid Robotics , Research Intern	Beijing, China 2026.03 – 2026.06
- Designed an embodied memory system to mitigate temporal inconsistency and memory drift in long-horizon video generation. - Built a World Action Model on top of the memory system to improve planning and generation consistency for long-horizon robotic tasks. - Investigated joint world-model pretraining on egocentric human-hand and robot data. - Built separate data-validation and cleaning pipelines for both data sources and integrated them into the organization’s existing data-processing workflow.	
Simplexity Robotics (Joint Lab with PKU) , Research Intern	Beijing, China 2025.09 – 2026.02
- Optimized whole-body control and mobile-manipulation coupling within a latent-reasoning fast-slow architecture; incorporated joint-torque proprioception and Human-in-the-Loop reinforcement learning for real-world robustness. - Contributed substantially to two generations of lightweight egocentric hand-data collection gloves. Both used a ZED 2i primary view, dual wrist-mounted Insta360 GO 3S	

cameras, and a Unity application for visualization and three-camera timestamp synchronization.

- Gen 1 used magnetic wrist tracking and multiple IMUs for finger capture; Gen 2 replaced magnetic tracking with SLAM and reduced the IMU count. Motion retargeting and accuracy validation preserved finger-joint localization while reducing cost and mitigating wrist-tracker dropouts.
- Incorporated multi-task, multi-view human-hand data into VLA pretraining and downstream training, improving out-of-distribution generalization.
- Deployed Galaxea R1 Lite and Franka Emika Panda robots end to end, spanning real-robot data collection and cleaning, model training, testing, and validation; supported bring-up of the in-house i7 Pro mobile-base and dual-arm system.
- Designed and fabricated custom 3D-printed robot components for rapid structural iteration.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern, Embodied Multimodal LLM Research Center

Beijing, China
2024.08 – 2025.08

- Co-built RoboBrain 1.0/2.0 to improve LLM/VLM spatial understanding and temporal modeling.
- Co-developed **AC-DiT**, an end-to-end VLA method that achieved SOTA performance across multiple mobile-manipulation benchmarks.
- Contributed to **RoboMIND (RSS 2025)**, comprising 107K demonstrations across 4 robot embodiments, 479 tasks, and 96 object classes, plus 5K annotated real-world failure demonstrations and an Isaac Sim digital-twin environment.
- Performed hardware bring-up and real-robot data collection for AgileX Mobile ALOHA platforms and a RealMan RM65-B-V compound lifting mobile manipulator.
- Built datasets and benchmarks across SAPIEN, MuJoCo, CoppeliaSim, and Isaac Sim; developed VR/SpaceMouse teleoperation pipelines and led construction of 100+ objects and 10+ scenes as cross-simulator digital assets.

Research Highlights

AC-DiT: Adaptive Coordination Diffusion Transformer for Mobile Manipulation

NeurIPS 2025

Co-developed an end-to-end mobile-manipulation VLA with coarse-to-fine action generation; led core implementation and simulation/real-robot evaluation.

LaST-HD: Learning Latent Physical Reasoning from Scalable Human Data for Robot Manipulation

arXiv 2026
Technical Report

Aligned unpaired human-hand and robot demonstrations in a shared latent physical-reasoning space; combined an action-conditioned world model, OOL Glove, and progressive mixed training for cross-embodiment transfer.

HDR: Hierarchical Denoising for Multi-Step Visual Reasoning and Robotic World Action Modeling

RSS 2026 Workshop
Best Paper

Contributed to a coarse-to-fine latent-tree framework for globally consistent multi-step visual reasoning; improved success over streaming baselines, ran 54.2x faster than bidirectional diffusion, and transferred to physical interaction.

RoboMIND: Benchmark on Multi-embodiment Intelligence Normative Data for Robot Manipulation

RSS 2025

Contributed to a multi-embodiment teleoperation dataset and benchmark with 107K demonstrations, 479 tasks, 96 object classes, and 4 robot embodiments.

LaST-R1: Reinforcing Robotic Manipulation via Adaptive Physical Latent Reasoning

arXiv 2026
Technical Report

Contributed to LAPO, an RL post-training framework jointly optimizing latent reasoning and action generation; achieved 99.9% average LIBERO success and up to 22.5% improvement over supervised fine-tuning across four real-world tasks.

EchoArena: Learning World Models for Reliable VLA Policy Evaluation

CVPR 2026
Workshop

Built an evaluator-centric world model that ranks five VLA policies across ten tasks and three embodiments against real-world outcomes.

Chain of Thought Prompt Tuning in Vision Language Models

arXiv 2023

Contributed to a Chain-of-Thought prompt-tuning method for VLMs that improves generalization, transfer, and reasoning across image classification, retrieval, and visual question answering.

Group-of-Latents: Perceptual Video Compression at Extreme Bitrates via Masked Latent Generative Modeling

ACM MM 2026

Contributed to causal latent video compression separating intra-frame perceptual anchors from inter-frame synthesis, achieving leading fidelity below 0.005 bpp with temporal consistency.

Implicit Neural Image Field for Biological Microscopy Image Compression

Nature
Computational Science

Contributed to an adaptive implicit-neural-representation pipeline supporting arbitrary-shape microscopy images and pixel-level decompression, with controllable compression up to 512x.

Open-Source Projects

CAMEL: Communicative Agents for Mind Exploration of LLM Society [↗](#)

Contributor | 17.6k stars

Designed Agent Tools and prompting schemes for Zero-shot and One-shot embodied task execution on RoboTwin, outperforming representative VLA baselines on unseen tasks and rapid adaptation.

AC-DiT: Adaptive Coordination Diffusion Transformer [↗](#)

NeurIPS 2025 | 49 stars

Implemented and engineered core multimodal-input and two-stage action-generation modules, together with simulation and real-robot evaluation scripts and configurations.

Awesome-World-Models [↗](#)

Maintainer | 2.0k stars

Maintain a structured research index spanning video prediction, action-conditioned dynamics, embodied AI, and reinforcement learning with world models.

Skills

- **Agent systems and long-horizon task solutions:** VLA/WAM design, training, evaluation, and deployment; fast-slow hierarchies, world models, and memory augmentation for robust long-horizon execution.
- **Robot systems and real-world deployment:** Hands-on with AgileX Mobile ALOHA, RealMan RM65-B-V, Galaxea R1 Lite, Franka Panda, and Simplexity i7 Pro; hardware bring-up, teleoperation, data pipelines, and deployment.
- **Robot simulation and evaluation:** SAPIEN, MuJoCo, CoppeliaSim, and Isaac Sim; cross-simulator datasets, digital assets, benchmarks, and VR/SpaceMouse teleoperation pipelines.
- **Human data collection and cross-embodiment transfer:** Co-designed two generations of egocentric gloves integrating magnetic/IMU/SLAM sensors, synchronized cameras, Unity collection software, motion retargeting, and human-to-robot transfer.
- **Engineering and research toolchain:** Proficient in Python, C/C++, PyTorch, and ROS/ROS2; experienced in research-code engineering, reproduction, system integration, and deployment.

Honors and Awards

University-level Outstanding Student; University-level Outstanding Communist Youth League Member; Grand Prize, Academic Competition Scholarship; First Prize, Academic Excellence Scholarship; First Prize, Innovation and Entrepreneurship Scholarship; Second Prize, Student Work Scholarship.

Additional Information

Patent applications

- Real-time collaborative multi-camera analysis for specified targets.
- Joint analysis using fixed ground cameras and mobile aerial drones.

Software copyrights

- Integrated ground-air video retrieval and trajectory playback.
- Integrated ground-air real-time and offline pedestrian/vehicle tracking.
- Integrated ground-air intelligent surveillance platform.

Leadership and interests: former class monitor and Youth League secretary at PKU's National Engineering Research Center of Visual Technology; former student journalist at Beihang; badminton and photography.